

Collision Ideals and Off-Diagonal Sheets

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Abstract

We study the scheme of collisions of a polynomial self-map—pairs of points with the same image—and describe its off-diagonal geometry through boundary-supported inertia and Galois-closure sheets. For a polynomial map $F = (F_1, \dots, F_n): \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, define in $S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$ the collision and diagonal ideals

$$I_R = (F_i(x) - F_i(y))_{i=1}^n, \quad I_{\Delta} = (x_i - y_i)_{i=1}^n.$$

The image of I_{Δ} in the collision algebra $C_F := S/I_R$ is the obstruction ideal $\text{Obs}(F) := I_{\Delta}/I_R$; it records the failure of the collision scheme $\text{Spec } C_F$ to coincide with the diagonal. Its vanishing characterizes polynomial automorphisms:

$$\text{Obs}(F) = 0 \iff F \text{ is a polynomial automorphism.}$$

In dimension two we separate two finiteness targets. Planar boundary coherence asks that the boundary local cohomology $H_{\partial X}^1(\overline{X}, \mathcal{O}_{\overline{X}})$ be finite; it forces $\partial X = \emptyset$, hence makes the Keller map finite étale and therefore an automorphism. The weaker planar ramification rigidity target asks that $\Omega_{A_N/B}$ have finite length. On the common Galois normalization $Z = \text{Spec } A_N$, we construct the pulled-back affine opens U_g , their canonical boundary ideals $\mathfrak{j}_g = I_{A_N}(Z \setminus U_g)$, and the fixed-locus ideals $\mathfrak{a}_C$. The height-one dictionary identifies moving-sheet coverage and finite-group boundary separation as exact criteria for ramification rigidity, after which purity and finite-étale rigidity give $N = L = K$ and collision vanishing. Although $\overline{A}$ and A_N are already finite over B , neither of these two stronger finiteness targets follows formally. Universal validity of either target is equivalent to the two-dimensional Jacobian conjecture and is not proved here. For the separable nonnormal cubic class in dimension three, let

$$K = \mathbb{C}(F_1, F_2, F_3), \quad L = \mathbb{C}(x_1, x_2, x_3),$$

and let N be the normal closure of L/K . Then

$$L \otimes_K L \simeq L \times N, \quad \text{Gal}(N/K) \simeq S_3.$$

The off-diagonal factor N is the ordered-root Galois-closure sheet. Consequently, $\text{Obs}(F) \neq 0$. The algebraic construction, the pulled-back boundary dictionary, and the conditional coverage/separation endgame are formalized in Lean 4. The boundary-coherence implication and the finite-length support bridge remain explicit mathematical inputs, as do the conjectural and external geometric statements.

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1 Introduction

The Jacobian conjecture asks whether a polynomial self-map of affine space with constant nonzero Jacobian determinant must be a polynomial automorphism. Dimension two remains open. Rather than treating injectivity as an external property of the map, this paper studies the scheme of pairs with equal image and asks when its off-diagonal part disappears.

There are two distinct outcomes. In the planar case, the paper does not prove the Jacobian conjecture. It organizes the remaining problem around two precise targets:

$$\begin{aligned} \text{PBC}(F) &: \Longleftrightarrow H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) \text{ is a finite } \overline{A}\text{-module,} \\ \text{PRR}(F) &: \Longleftrightarrow \Omega_{A_N/B} \text{ has finite length over } A_N. \end{aligned}$$

Planar boundary coherence $\text{PBC}(F)$ is the stronger properness route:

$$\text{PBC}(F) \implies \partial X = \emptyset \implies f_F \text{ finite étale} \implies F \text{ an automorphism.}$$

Planar ramification rigidity $\text{PRR}(F)$ is the weaker codimension-one route:

$$\text{PRR}(F) \implies Z/Y \text{ finite étale} \implies N = K \implies L = K \implies q_F = 0 \implies I_R = I_{\Delta}.$$

The second route is controlled by an exact codimension-one obstruction on the Galois normalization. A ramified divisor E is eliminated as soon as one conjugate sheet moved by I_E still contains its generic point. In coordinates this is the conjugate-integrality condition

$$I_E \neq 1 \implies \exists g \in G, \quad I_E \not\subseteq gHg^{-1}, \quad w_E(g(x_1)), w_E(g(x_2)) \geq 0.$$

We call this planar moving-sheet coverage. Its ideal-theoretic form says that the fixed locus of a prime-order subgroup and the boundaries of all the sheets it moves have no common height-one component. Establishing this universally—equivalently, establishing planar ramification rigidity universally—is equivalent to $J\mathcal{C}(2)$, so the reformulation locates the missing theorem without assuming it.

Only after all height-one points have been shown unramified do purity and finite-étale rigidity enter: they make the normal closure trivial and collapse the collision relation to the diagonal. This order is essential to the argument. The paper also develops boundary divisor-class and principal-parts criteria that may imply coverage under additional hypotheses; these are sufficient routes, not proofs of universal coverage.

The normalization rings $\overline{A}$ and A_N are finite B -modules from the outset. This makes $\overline{X}$ and Z affine varieties: they are integral affine schemes of finite type over $\mathbb{C}$. It does not make A finite over B , make boundary local cohomology finite, or make $\Omega_{A_N/B}$ finite length. Likewise, the vanishing of ordinary higher quasi-coherent cohomology on an affine scheme does not imply the vanishing or finiteness of local cohomology with support in its deleted boundary.

In the separable nonnormal cubic class in dimension three, the opposite phenomenon is visible generically: the off-diagonal factor is the ordered-root S_3 -normal-closure sheet. This gives a structural comparison with the planar target while keeping its assumptions explicit.

Section 2 develops collision ideals and the dimension-independent normalization diagram. Section 3 constructs the planar pulled-back boundaries, proves the valuation dictionary, and states the equivalent coverage and separation criteria before applying the endgame. Section 4 treats the cubic S_3 class.

2 General construction

The organizing question is how the failure of a polynomial self-map to be one-to-one is encoded by the part of its fiber square lying off the diagonal. The collision ideal first isolates that part scheme-theoretically; the normalization diagram then describes it through conjugate sheets and inertia. These constructions are dimension-independent. Section 3 specializes them to planar Keller maps and locates the obstruction at the deleted normalization boundary, while Section 4 identifies the generic off-diagonal sheet in the separable nonnormal cubic class with an S_3 -normal closure. The two specializations respectively describe when the collision scheme collapses to the diagonal and what its surviving generic sheet looks like when it does not.

2.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n): X = \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map F is Keller if

$$\det JF \in \mathbb{C}^\times.$$

Define

$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \quad I_\Delta = (x_i - y_i)_{i=1}^n, \quad \text{Obs}(F) := I_\Delta / I_R(F).$$

The quotient $\text{Obs}(F)$ is naturally an ideal of the collision ring $C_F := S/I_R(F)$; we call it the obstruction ideal. For a fixed map F , we abbreviate $I_R(F)$ to I_R .

Proposition 2.1 (Canonical containment). *For every polynomial map F ,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $H \in \mathbb{C}[x_1, \dots, x_n]$, the difference $H(x) - H(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on H , or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i . $\square$

Let $A := \mathbb{C}[x_1, \dots, x_n]$. Diagonal evaluation induces a surjective ring map

$$\bar{\mu}_F: C_F \longrightarrow A, \quad [s]_{I_R} \longmapsto s(x, x).$$

Proposition 2.2 (The diagonal kernel). *As ideals of C_F ,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta / I_R.$$

Consequently,

$$\ker(\bar{\mu}_F) = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof. The kernel of diagonal evaluation $S \rightarrow A$ is I_Δ . Passing through S/I_R therefore identifies the displayed kernel with I_Δ / I_R . Since $I_R \subseteq I_\Delta$,

$$I_\Delta / I_R = 0 \iff I_R = I_\Delta.$$

$\square$

2.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let $X = \text{Spec } A$, let $Y = \text{Spec } B$, and let

$$f_F: X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y: A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F . Their common restriction equips C_F with a B -algebra structure.

Theorem 2.3 (Fiber-product interpretation). *There is a canonical B -algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \quad R_F := \text{Spec } C_F \cong X \times_Y X.$$

Under the first equivalence, $\bar{\mu}_F$ is tensor multiplication

$$A \otimes_B A \longrightarrow A, \quad a \otimes a' \longmapsto aa'.$$

Its dual morphism is the diagonal

$$\Delta_{X/Y}: X \longrightarrow X \times_Y X.$$

Proof. A map $C_F \rightarrow T$ to a commutative $\mathbb{C}$ -algebra T is equivalent to a pair of maps $f, g: A \rightrightarrows T$ satisfying

$$f(F_i) = g(F_i) \quad (1 \leq i \leq n).$$

Since the F_i generate B , this is precisely the universal property of $A \otimes_B A$. Applying Spec to this equivalence and using

$$\text{Spec}(A \otimes_B A) \cong \text{Spec } A \times_{\text{Spec } B} \text{Spec } A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. $\square$

$$R_F(\mathbb{C}) = \{(u, v) \in X(\mathbb{C}) \times X(\mathbb{C}) : F(u) = F(v)\}.$$

Inside

$$X \times X = \text{Spec } S,$$

the ideals I_R and I_Δ define, respectively, $X \times_Y X$ and the image of $\Delta_{X/Y}$. The contravariance of $I \mapsto V(I)$ therefore gives the equivalences

$$I_R \subseteq I_\Delta \iff \Delta_{X/Y}(X) \subseteq X \times_Y X$$

and

$$I_R = I_\Delta \iff X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes of $X \times X$. Thus the algebraic and geometric gaps are two descriptions of the same scheme-theoretic containment.

2.3 The off-diagonal factor and secant projectors

For every polynomial map F , define

$$I_{\text{off}} := I_R : I_\Delta, \quad C_F^\circ := S/I_{\text{off}}, \quad R_F^\circ := \text{Spec } C_F^\circ.$$

This definition does not require the diagonal to be clopen.

Corollary 2.4 (Off-diagonal emptiness criterion). *For every polynomial map F , the following are equivalent:*

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof.

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff I_R : I_\Delta = S.$$

Moreover,

$$I_R : I_\Delta = S \iff I_\Delta \subseteq I_R \iff I_R = I_\Delta.$$

The equivalence with $\text{Obs}(F) = 0$ is Proposition 2.2. $\square$

The idempotent mechanism behind a clopen diagonal is purely commutative-algebraic. Let C be a commutative ring, $J \subseteq C$ an ideal, $\delta \in C$, and $c \in C^\times$. Suppose

$$\delta J = 0, \quad \delta - c \in J.$$

Define

$$q := 1 - c^{-1}\delta.$$

Theorem 2.5 (Abstract off-diagonal projector). *The element q is idempotent and*

$$J = Cq, \quad \text{Ann}_C(J) = C(1 - q).$$

In particular,

$$J = 0 \iff q = 0.$$

Proof. The congruence $\delta \equiv c \pmod{J}$ gives $q \in J$. For every $j \in J$,

$$qj = j - c^{-1}\delta j = j.$$

Taking $j = q$ proves $q^2 = q$; the same identity gives $J = Cq$. Finally,

$$aq = 0 \iff a = a(1 - q),$$

so $\text{Ann}_C(J) = \text{Ann}_C(Cq) = C(1 - q)$. □

Whenever the theorem is applied with

$$C = C_F, \quad J = \text{Obs}(F),$$

the identities $J = C_F q$ and $C_F/J \cong A$ give

$$C_F \cong A \times C_F/J(1 - q).$$

Thus the first factor is the diagonal and the second is its off-diagonal complement.

2.4 Generic fibers and marked-root decomposition

Let F be dominant and generically finite, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

Let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Theorem 2.6 (Generic collision bridge). *The map θ is an isomorphism. Consequently, there is a K -algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ \sim \downarrow & & \downarrow \theta \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \xrightarrow{\sim} U^{-1}A, \quad \frac{b}{u} \otimes a \mapsto \frac{ba}{u}.$$

Since $A = \mathbb{C}[x_1, \dots, x_n]$ is a domain and $B \subseteq A$,

$$B \setminus \{0\} \subseteq A \setminus \{0\}.$$

Hence

$$U^{-1}A \hookrightarrow \text{Frac}(A) = L, \quad \frac{a}{u} \mapsto \frac{a}{u}.$$

Composing this injection with the displayed isomorphism gives θ , so θ is injective.

Since F is generically finite, L/K is a finite extension. The image of $U^{-1}A$ in L is

$$K[x_1, \dots, x_n].$$

Since each x_i is algebraic over K , this K -domain is finite-dimensional over K , hence is a field. Since it contains A , it contains $\text{Frac}(A) = L$; the reverse inclusion is already given by its embedding into L . Therefore

$$U^{-1}A = L,$$

and θ is an isomorphism.

By Theorem 2.3, identify C_F with $A \otimes_B A$. There is a K -algebra isomorphism

$$\beta: K \otimes_B (A \otimes_B A) \xrightarrow{\sim} (K \otimes_B A) \otimes_K (K \otimes_B A)$$

defined on pure tensors by

$$\beta(\lambda \otimes (a \otimes a')) = (\lambda \otimes a) \otimes (1 \otimes a').$$

Its inverse is

$$(\lambda \otimes a) \otimes (\lambda' \otimes a') \mapsto \lambda \lambda' \otimes (a \otimes a').$$

The balancing relations over B and K show that these formulas are well-defined, and the two displayed maps are mutually inverse. Composing β with $\theta \otimes_K \theta$ gives

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L.$$

It remains to identify the diagonal map. Under $C_F \cong A \otimes_B A$, the homomorphism $\bar{\mu}_F$ is multiplication. Therefore, on a pure tensor,

$$\begin{aligned} (\mu_L \circ \Phi_F)(\lambda \otimes (a \otimes a')) &= (\lambda a) a' = \lambda a a', \\ (\theta \circ (1 \otimes \bar{\mu}_F))(\lambda \otimes (a \otimes a')) &= \theta(\lambda \otimes a a') = \lambda a a'. \end{aligned}$$

Pure tensors generate $K \otimes_B C_F$, so the two homomorphisms are equal. This proves the commutative square. $\square$

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α , and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Theorem 2.7 (Marked-root tensor decomposition). *Give $L \otimes_K L$ its L -algebra structure through the left tensor factor. There is an L -algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

under which tensor multiplication $L \otimes_K L \rightarrow L$ is projection onto the first factor.

Proof. The power-basis presentation

$$L \cong K[T]/(m_\alpha)$$

and base change from K to L give an L -algebra isomorphism

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \quad 1 \otimes \alpha \mapsto T.$$

Since m_α is separable, $m'_\alpha(\alpha) \neq 0$. Differentiating

$$m_\alpha(T) = (T - \alpha)h_\alpha(T)$$

and evaluating at $T = \alpha$ gives

$$h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0.$$

Thus $T - \alpha$ does not divide h_α , so these two polynomials are coprime in $L[T]$. The Chinese remainder theorem now gives

$$L[T]/(m_\alpha) \xrightarrow{\sim} L[T]/(T - \alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim} L \times L[T]/(h_\alpha).$$

Under the first isomorphism above, a pure tensor $\ell \otimes p(\alpha)$, with $p \in K[T]$, corresponds to the class of $\ell p(T)$. Tensor multiplication sends it to $\ell p(\alpha)$, which is evaluation at $T = \alpha$. Under the Chinese remainder isomorphism, this evaluation map is projection onto the factor $L[T]/(T - \alpha) \cong L$. Hence tensor multiplication is the first projection. $\square$

Together with Theorem 2.6, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

2.5 Normalized covers and inertia

Let L/K be finite and separable, and let N/K be its normal closure, with a fixed embedding $L \hookrightarrow N$. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Proposition 2.8 (Conjugate-embedding action). *Restriction induces a bijection*

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \mapsto g|_L.$$

The induced action of G on G/H is faithful; equivalently,

$$\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

Proof. For $g_1, g_2 \in G$,

$$g_1|_L = g_2|_L \iff g_2^{-1}g_1 \in H \iff g_1H = g_2H.$$

Every K -embedding $L \hookrightarrow N$ extends to an element of G , so restriction gives the stated bijection. Its action kernel is

$$\ker(G \curvearrowright G/H) = \bigcap_{g \in G} gHg^{-1} = \text{core}_G(H).$$

If $\sigma \in \text{core}_G(H)$, then

$$\sigma|_{g(L)} = \text{id}_{g(L)} \quad (g \in G).$$

Since N is the normal closure,

$$N = K(g(L) : g \in G),$$

and therefore $\sigma = \text{id}_N$. □

Let

$$Y = \text{Spec } B, \quad \overline{X} = \text{Norm}_L(Y), \quad Z = \text{Norm}_N(Y).$$

The marked embedding $L \hookrightarrow N$ induces a canonical commutative triangle

$$Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, \quad \nu = \bar{\nu} \circ \pi : Z \longrightarrow Y.$$

Suppose that $\bar{\nu}$ and ν are finite, and fix an open immersion over Y

$$j : X = \text{Spec } A \hookrightarrow \overline{X}$$

with $\bar{\nu} \circ j = f_F$. Denote this diagram and its deleted boundary by

$$\mathcal{D} = \left(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, j : X \hookrightarrow \overline{X} \right), \quad \partial X = \overline{X} \setminus j(X).$$

Fix $E \in Z^{(1)}$. Let w_E be its divisorial valuation, let $b = \nu(E)$, and let $v_b = w_E|_K$. Let $D_E \leq G$ and $I_E \trianglelefteq D_E$ be its decomposition and inertia groups. Let

$$\mathcal{Q}_E := D_E \backslash G/H.$$

For $\omega = D_E g H \in \mathcal{Q}_E$, let

$$u_\omega = (g^{-1}w_E)|_L, \quad \text{cent}(E, \omega) = \text{cent}_{\overline{X}}(u_\omega),$$

and let

$$i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

Proposition 2.9 (Conjugate divisors and double cosets). *The assignments above are well-defined, and restriction of valuations induces a bijection*

$$\mathcal{Q}_E \xrightarrow{\sim} \{\text{prime divisors of } \overline{X} \text{ above } b\}, \quad \omega \longmapsto \text{cent}(E, \omega).$$

Proof. The group G acts transitively on the extensions of v_b to N , and the stabilizer of w_E is D_E . Restricting such a valuation to $L = N^H$ identifies precisely those extensions whose indices differ on the right by an element of H . The resulting orbits are therefore indexed by $D_E \backslash G/H$, giving the stated bijection.

It remains to check independence of representatives. If $D_E g' H = D_E g H$, then $g' = dgh$ for some $d \in D_E$ and $h \in H$. The element d fixes w_E , while h fixes L pointwise, so

$$(g'^{-1}w_E)|_L = (g^{-1}w_E)|_L.$$

Moreover, $I_E \trianglelefteq D_E$, and conjugation by d identifies

$$I_E \cap g' H g'^{-1} \quad \text{with} \quad I_E \cap g H g^{-1}.$$

Thus u_ω , $\text{cent}(E, \omega)$, and $i_E(\omega)$ are well-defined. $\square$

We use the notation

$$e(\text{cent}(E, \omega)/b) := e(u_\omega/v_b).$$

Since the residue characteristic is zero,

$$\text{Ram}^{(1)}(Z/Y) := \{E \in Z^{(1)} : e(w_E/v_b) > 1\} = \{E \in Z^{(1)} : I_E \neq 1\}.$$

Proposition 2.10 (Detection of inertia on a conjugate sheet). *If $I_E \neq 1$, then there is an $\omega \in \mathcal{Q}_E$ with*

$$i_E(\omega) > 1.$$

Proof. Suppose, to the contrary, that $i_E(\omega) = 1$ for every $\omega \in \mathcal{Q}_E$. For each $g \in G$,

$$1 = i_E(D_E g H) = [I_E : I_E \cap g H g^{-1}],$$

so

$$I_E = I_E \cap g H g^{-1} \subseteq g H g^{-1} \quad \text{for every } g \in G.$$

Consequently,

$$I_E \subseteq \bigcap_{g \in G} g H g^{-1} = \text{core}_G(H) = 1,$$

contrary to $I_E \neq 1$. $\square$

Proposition 2.11 (Intermediate ramification index). *For $\omega = D_E g H \in \mathcal{Q}_E$,*

$$e(\text{cent}(E, \omega)/b) = i_E(\omega) = [I_E : I_E \cap g H g^{-1}].$$

Proof. Let w_E and v_b be the divisorial valuations associated with E and b , and let

$$w_g = g^{-1}w_E, \quad u_\omega = w_g|_L.$$

The valuation u_ω has center $\text{cent}(E, \omega)$ on $\overline{X}$. Since the residue fields have characteristic zero,

$$e(w_g/v_b) = |I_E|.$$

The inertia group of w_g over u_ω is

$$g^{-1}I_E g \cap H,$$

and hence

$$e(w_g/u_\omega) = |g^{-1}I_E g \cap H| = |I_E \cap gHg^{-1}|.$$

Multiplicativity of ramification indices in the field tower

$$K \subseteq L \subseteq N$$

therefore gives

$$e(u_\omega/v_b) = \frac{e(w_g/v_b)}{e(w_g/u_\omega)} = [I_E : I_E \cap gHg^{-1}],$$

which is the asserted formula. $\square$

Proposition 2.12 (Visibility-to-boundary). *Suppose that $X \rightarrow Y$ is étale. For $E \in Z^{(1)}$ and $\omega \in \mathcal{Q}_E$,*

$$i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X.$$

Proof. Suppose that $\text{cent}(E, \omega) \in j(X)$, and let $x \in X$ be the corresponding point under the open immersion j . The local homomorphism

$$\mathcal{O}_{Y,b} \longrightarrow \mathcal{O}_{X,x}$$

is étale, so $e(\text{cent}(E, \omega)/b) = 1$. Proposition 2.11 now gives

$$i_E(\omega) = e(\text{cent}(E, \omega)/b) = 1.$$

Taking the contrapositive gives the result. $\square$

Definition 2.13 (Hidden-inertia locus).

$$\mathcal{H}(\mathcal{D}) := \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E, i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

Thus $E \in \mathcal{H}(\mathcal{D})$ precisely when nontrivial inertia is visible on a conjugate sheet but every visible center belongs to the deleted boundary.

Proposition 2.14 (Divisorial ramification is hidden from the étale sheet). *Suppose that N/K is the normal closure of L/K , the residue characteristic is zero, and $X \rightarrow Y$ is étale. Then*

$$\mathcal{H}(\mathcal{D}) = \text{Ram}^{(1)}(Z/Y).$$

Equivalently, every divisorial ramification orbit has a conjugate sheet on which its inertia acts nontrivially, but every such center lies in the deleted boundary.

Proof. The inclusion $\mathcal{H}(\mathcal{D}) \subseteq \text{Ram}^{(1)}(Z/Y)$ is part of the definition. Conversely, let $E \in \text{Ram}^{(1)}(Z/Y)$. In residue characteristic zero, $I_E \neq 1$. Since N is the normal closure of L/K , Proposition 2.8 gives $\text{core}_G(H) = 1$, and Proposition 2.10 supplies an $\omega \in \mathcal{Q}_E$ with $i_E(\omega) > 1$. For every $\omega' \in \mathcal{Q}_E$, Proposition 2.12 gives

$$i_E(\omega') > 1 \implies \text{cent}(E, \omega') \in \partial X.$$

These are exactly the two conditions defining $E \in \mathcal{H}(\mathcal{D})$. $\square$

2.6 Automorphism detection

Theorem 2.15 (Automorphism criterion). *For an arbitrary polynomial self-map $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, the following are equivalent:*

- (1) F is a polynomial automorphism;
- (2) F is injective on $\mathbb{A}^n(\mathbb{C})$;
- (3) $I_R(F) = I_{\Delta}$;
- (4) $\text{Obs}(F) = 0$.

Proof. Suppose first that F has polynomial inverse $G = (G_1, \dots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i , so $I_{\Delta} \subseteq I_R$. Proposition 2.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_{\Delta}$, and let $\text{ev}_{(a,b)}: S \rightarrow \mathbb{C}$ be evaluation at (a, b) . If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\text{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_{\Delta} = I_R$ gives $a_i - b_i = 0$ for every i . Thus $a = b$, proving (3) $\Rightarrow$ (2). The Ax–Grothendieck theorem [Ax69] gives (2) $\Rightarrow$ (1). Finally, Proposition 2.2 gives (3) $\iff$ (4). $\square$

Corollary 2.16 (The Jacobian conjecture as obstruction vanishing). *For every $n \geq 1$, the n -dimensional Jacobian conjecture is equivalent to each of the statements*

$$\forall F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^{\times} \implies \text{Obs}(F) = 0,$$

and

$$\forall F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^{\times} \implies I_R(F) = I_{\Delta}.$$

Proof. Apply Theorem 2.15 to every Keller map. Its equivalent conditions (1), (3), and (4) give the two displayed formulations. $\square$

2.7 Generic degree one

Corollary 2.17 (Generic degree-one descent). *If*

$$L = K \quad \text{and} \quad A \text{ is flat over } B,$$

then

$$\text{Obs}(F) = 0.$$

Proof. When $L = K$, Theorem 2.6 identifies the base-changed diagonal with tensor multiplication:

$$\ker(1 \otimes \bar{\mu}_F) \cong \ker\left(K \otimes_K K \xrightarrow{\mu_K} K\right) = 0.$$

The fiber-product identification of Theorem 2.3 is

$$C_F \xrightarrow{\sim} A \otimes_B A.$$

Since A is flat over B , the tensor product $A \otimes_B A$, and therefore C_F , is flat over B . The localization map

$$\iota_C: C_F \hookrightarrow K \otimes_B C_F$$

is therefore injective, and

$$\iota_C(\ker \bar{\mu}_F) \subseteq \ker(1 \otimes \bar{\mu}_F) = 0.$$

Thus $\ker \bar{\mu}_F = 0$. Proposition 2.2 identifies

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta/I_R,$$

so $\text{Obs}(F) = 0$. □

Lean correspondence. `CollisionIdeals/CollisionDiagonal.lean`, `CollisionIdeals/OffDiagonalScheme.lean`, `CollisionIdeals/Secant.lean`, `CollisionIdeals/GenericCollisionDiagonal.lean`, `CollisionIdeals/KellerCollisionModel.lean`, and `CollisionIdeals/JacobianConjecture.lean`. The Lean form of Theorem 2.6 takes surjectivity of θ as a parameter; the proof above derives it from generic finiteness.

Remark (Classical complex $JC(2)$). For clarity, the traditional complex two-dimensional Jacobian conjecture is the statement

$$\forall F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF \in \mathbb{C}^\times \implies F \text{ is a polynomial automorphism.}$$

No normalization boundary, Galois-closure, coverage, separation, or module-finiteness condition is part of this definition. By Corollary 2.16, classical $JC(2)$ is equivalently the assertion that every such planar Keller map satisfies

$$\text{Obs}(F) = 0 \quad \text{or, equivalently,} \quad I_R(F) = I_\Delta.$$

The planar section introduces additional geometric formulations under their own names; their precise logical relation to classical $JC(2)$ is recorded only after those formulations have been developed.

3 Planar vanishing

We now specialize the dimension-independent collision, normalization, and inertia constructions of Section 2 to dimension two. The purpose of this section is to identify a concrete boundary obstruction, not to assume it away implicitly: the secant calculation separates the diagonal from the off-diagonal collision sheet, while the normalization calculation shows exactly how nontrivial inertia would have to escape every relevant conjugate affine sheet.

Let

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2$$

be a polynomial map satisfying $\det JF = c \in \mathbb{C}^\times$.

Set, from this point onward,

$$A = \mathbb{C}[x_1, x_2], \quad B = \mathbb{C}[P, Q] \subseteq A, \quad K = \mathbb{C}(P, Q), \quad L = \mathbb{C}(x_1, x_2),$$

and let $X = \text{Spec } A$, $Y = \text{Spec } B$.

Proposition 3.1 (Planar Keller geometry).

$$\mathbb{C}[u, v] \xrightarrow{\sim} B, \quad u \mapsto P, \quad v \mapsto Q.$$

Moreover,

$$X \xrightarrow{f_F} Y \text{ is étale,} \quad A \text{ is flat over } B, \quad [L : K] < \infty \text{ and } L/K \text{ is separable.}$$

Proof. The identity

$$dP \wedge dQ = \left(\frac{\partial P}{\partial x_1} \frac{\partial Q}{\partial x_2} - \frac{\partial P}{\partial x_2} \frac{\partial Q}{\partial x_1} \right) dx_1 \wedge dx_2 = c dx_1 \wedge dx_2 \neq 0$$

shows that P, Q are algebraically independent. Thus $B \cong \mathbb{C}[u, v]$ and $Y \cong \mathbb{A}_{\mathbb{C}}^2$. Moreover,

$$A \cong \mathbb{C}[u, v, U, V] / (P(U, V) - u, Q(U, V) - v)$$

as a $\mathbb{C}[u, v]$ -algebra, and its relative Jacobian determinant is $c \in A^\times$. The Jacobian criterion therefore makes $B \rightarrow A$ étale. In particular it is flat and quasi-finite. Since f_F is dominant and quasi-finite, its generic fiber is zero-dimensional, so $[L : K] < \infty$. The morphism remains étale at the generic point; hence the finite extension L/K is separable. $\square$

Proposition 3.2 (Finite planar Keller rigidity). *If the planar Keller morphism $f_F : \mathbb{A}_{\mathbb{C}}^2 \rightarrow \mathbb{A}_{\mathbb{C}}^2$ is finite, then F is a polynomial automorphism.*

Proof. Proposition 3.1 makes f_F étale. Thus a finite f_F is a connected finite étale cover of $\mathbb{A}_{\mathbb{C}}^2$. Finite-étale rigidity of the affine plane, Theorem 3.30, makes it an isomorphism. Its inverse is a morphism of affine schemes and hence is given by polynomials. $\square$

Definition 3.3 (Planar generic degree). The generic degree of F is

$$d_F := \deg_{\text{gen}}(F) := [L : K].$$

Because L/K is finite and separable, d_F is also the number of K -embeddings $L \hookrightarrow \overline{K}$, equivalently the number of geometric points in the generic fiber of f_F .

The inverse Jacobian makes the two-dimensional structure explicit. The polynomial derivations

$$\partial_P = c^{-1} \left(Q_{x_2} \frac{\partial}{\partial x_1} - Q_{x_1} \frac{\partial}{\partial x_2} \right), \quad \partial_Q = c^{-1} \left(-P_{x_2} \frac{\partial}{\partial x_1} + P_{x_1} \frac{\partial}{\partial x_2} \right)$$

satisfy

$$\partial_P(P) = 1, \quad \partial_P(Q) = 0, \quad \partial_Q(P) = 0, \quad \partial_Q(Q) = 1,$$

and form an A -basis of $\text{Der}_{\mathbb{C}}(A)$. They commute: their commutator kills K , hence extends to a K -derivation of the finite separable extension L/K , and is therefore zero. This gives an explicit planar polynomial frame on the affine sheet; the issue below is whether inertia-moving conjugate frames can all escape that sheet at the normalization boundary.

3.1 The planar secant projector

Let $\Delta = V(I_\Delta) \subseteq \text{Spec } S$. For $a \in S$, write $a|_\Delta$ for its image under $S \rightarrow S/I_\Delta$, and apply this notation entrywise to matrices. Let

$$\mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

For $H \in \mathbb{C}[x_1, x_2]$, let

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1}, \quad D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients belong to S . Let

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \quad M_F|_\Delta = JF.$$

Let $\delta_F = \det(M_F)$.

Proposition 3.4 (Planar secant annihilation). *The secant determinant satisfies*

$$\delta_F I_\Delta \subseteq I_R, \quad \delta_F|_\Delta = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by $\text{adj}(M_F)$ gives

$$\delta_F \mathbf{d} = \text{adj}(M_F) \mathbf{r}.$$

$$\mathbf{r} \in I_R^{\oplus 2}, \quad \text{adj}(M_F) \in M_2(S) \implies \text{adj}(M_F) \mathbf{r} \in I_R^{\oplus 2}.$$

Therefore

$$\delta_F \mathbf{d} \in I_R^{\oplus 2},$$

or equivalently,

$$\forall j \in \{1, 2\}, \quad \delta_F(x_j - y_j) \in I_R.$$

Hence $\delta_F I_\Delta \subseteq I_R$. On the diagonal,

$$(D_j F_i)|_\Delta = \frac{\partial F_i}{\partial x_j},$$

so $M_F|_\Delta = JF$ and $\delta_F|_\Delta = \det JF = c$. □

Let $\bar{\delta}_F$ be the class of δ_F in C_F , and define

$$q_F := 1 - c^{-1} \bar{\delta}_F.$$

Theorem 3.5 (Planar collision-sheet decomposition). *The collision scheme decomposes as*

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ, \quad C_F \cong A \times C_F^\circ.$$

The off-diagonal ideal satisfies

$$I_{\text{off}} = I_R + (\delta_F) = I_R : I_\Delta^\infty.$$

Proof. By Proposition 2.2,

$$J := \ker(\bar{\mu}_F) = I_\Delta / I_R = \text{Obs}(F).$$

Proposition 3.4 gives

$$\bar{\delta}_F J = 0, \quad \bar{\delta}_F - c \in J.$$

Theorem 2.5, applied to these two relations and $q_F = 1 - c^{-1}\bar{\delta}_F$, yields

$$q_F^2 = q_F, \quad J = C_F q_F, \quad \text{Ann}_{C_F}(J) = C_F(1 - q_F), \quad J = 0 \iff q_F = 0.$$

Because $1 - q_F = c^{-1}\bar{\delta}_F$,

$$\frac{I_R : I_\Delta}{I_R} = \text{Ann}_{C_F}(J) = C_F(1 - q_F) = C_F \bar{\delta}_F = \frac{I_R + (\delta_F)}{I_R}.$$

Hence

$$I_{\text{off}} = I_R : I_\Delta = I_R + (\delta_F).$$

Idempotence gives $J^m = C_F q_F = J$ for every $m \geq 1$. Therefore

$$\frac{I_R : I_\Delta^m}{I_R} = \text{Ann}_{C_F}(J^m) = \text{Ann}_{C_F}(J) = \frac{I_R : I_\Delta}{I_R}.$$

Thus

$$I_R : I_\Delta^\infty = \bigcup_{m \geq 1} (I_R : I_\Delta^m) = I_R : I_\Delta.$$

The idempotent q_F also gives a ring isomorphism

$$C_F \xrightarrow{\sim} C_F(1 - q_F) \times C_F q_F, \quad a \mapsto (a(1 - q_F), a q_F),$$

whose inverse is addition. The two factors satisfy

$$C_F(1 - q_F) \cong C_F / C_F q_F = C_F / J \xrightarrow[\bar{\mu}_F]{\sim} A$$

and

$$C_F q_F \cong C_F / C_F(1 - q_F) \cong S / (I_R + (\delta_F)) = C_F^\circ.$$

Thus $C_F \cong A \times C_F^\circ$. Since the spectrum of a product is the disjoint union of the two spectra, this isomorphism dualizes to

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ.$$

□

Corollary 2.4 states that

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

The preceding theorem identifies

$$C_F^\circ = S / (I_R + (\delta_F)),$$

and Theorem 2.5 gives $\text{Obs}(F) = 0 \iff q_F = 0$. Hence

$$q_F = 0 \iff I_R + (\delta_F) = S \iff R_F^\circ = \emptyset \iff I_R = I_\Delta.$$

In particular, the secant determinant cuts out the off-diagonal factor:

$$R_F^\circ = \text{Spec}(S / (I_R + (\delta_F))).$$

3.2 A criterion for ramification rigidity: boundary principal parts

We first record an auxiliary local-cohomological construction that does not require a Galois closure. It will later provide sufficient criteria for the single codimension-one target $\text{PRR}(F)$; it is not a third planar target. Let R be a normal Noetherian domain of dimension two, write $W = \text{Spec } R$, and let

$$U = W \setminus V(\mathfrak{j})$$

be a dense open subset, where $\mathfrak{j} \subseteq R$ is a nonzero radical ideal. Define the boundary principal-parts module by

$$\mathcal{P}_{\mathfrak{j}}(R) := H_{\mathfrak{j}}^1(R).$$

The local-cohomology sequence for $U \subseteq W$ gives

$$0 \longrightarrow R \longrightarrow \Gamma(U, \mathcal{O}_U) \longrightarrow \mathcal{P}_{\mathfrak{j}}(R) \longrightarrow 0,$$

because $H_{\mathfrak{j}}^0(R) = 0$. Thus $\mathcal{P}_{\mathfrak{j}}(R)$ records the rational functions that are regular away from the boundary but do not extend across it [The26g].

Proposition 3.6 (Height-one boundary detector). *For $\mathfrak{p} \in \text{Spec}^{(1)} R$,*

$$\mathcal{P}_{\mathfrak{j}}(R)_{\mathfrak{p}} \cong \begin{cases} 0, & \mathfrak{j} \not\subseteq \mathfrak{p}, \\ \text{Frac}(R_{\mathfrak{p}})/R_{\mathfrak{p}}, & \mathfrak{j} \subseteq \mathfrak{p}. \end{cases}$$

Proof. Local cohomology commutes with localization. If $\mathfrak{j} \not\subseteq \mathfrak{p}$, then $\mathfrak{j}R_{\mathfrak{p}} = R_{\mathfrak{p}}$, so the localization vanishes. Otherwise $R_{\mathfrak{p}}$ is a discrete valuation ring and the nonzero ideal $\mathfrak{j}R_{\mathfrak{p}}$ has radical $\mathfrak{p}R_{\mathfrak{p}}$. The one-element Čech complex then identifies its first local cohomology with $\text{Frac}(R_{\mathfrak{p}})/R_{\mathfrak{p}}$. $\square$

Ramification can now be imposed as a second, independent filter. Suppose that R is finite flat and generically separable over a regular two-dimensional domain B , and let $\mathfrak{d} = \mathfrak{D}_{R/B}$ be the different. Set

$$\mathcal{P}_{\mathfrak{j}}^{\text{ram}}(R/B) := H_{\mathfrak{d}}^0(\mathcal{P}_{\mathfrak{j}}(R)).$$

This is the part of the boundary principal-parts module supported on the ramification locus. The spectral sequence for successive supports, together with $H_{\mathfrak{j}}^0(R) = 0$, gives a canonical identification

$$\mathcal{P}_{\mathfrak{j}}^{\text{ram}}(R/B) \cong H_{\mathfrak{d}+\mathfrak{j}}^1(R).$$

Proposition 3.7 (Uniform-annihilator criterion). *If there is an $m > 0$ such that*

$$(\mathfrak{d} + \mathfrak{j})^m H_{\mathfrak{d}+\mathfrak{j}}^1(R) = 0,$$

then

$$\mathcal{P}_{\mathfrak{j}}^{\text{ram}}(R/B) = 0.$$

Equivalently, the ramification locus and $V(\mathfrak{j})$ have no common height-one component.

Proof. The uniform annihilator makes $H_{\mathfrak{d}+\mathfrak{j}}^1(R)$ a finite R -module [The26d]. If a height-one prime $\mathfrak{p}$ contained $\mathfrak{d} + \mathfrak{j}$, Proposition 3.6, applied to that sum, would identify the localization with $\text{Frac}(R_{\mathfrak{p}})/R_{\mathfrak{p}}$, which is not finite over the discrete valuation ring $R_{\mathfrak{p}}$. Hence the support has codimension at least two. A normal Noetherian surface is Cohen–Macaulay, so depth gives $H_{\mathfrak{d}+\mathfrak{j}}^1(R) = 0$. $\square$

The proposition is a rigidity criterion, not an automatic vanishing statement. Its advantage is that it replaces an unbounded collection of possible pole orders by one concrete target: a uniform annihilator for the ramification-supported boundary principal parts. We now apply this general construction to the conjugate affine sheets supplied by the normalization diagram of Section 2.

3.3 Galois specialization and conjugate coordinates

Fix an algebraic closure $\overline{K} \supseteq L$, and let

$$\begin{aligned} N &:= K(\sigma(L) : \sigma \in \text{Hom}_K(L, \overline{K})), & K &\subseteq L \subseteq N \subseteq \overline{K}, \\ \overline{A} &:= \{a \in L : a \text{ is integral over } B\}, & A_N &:= \{a \in N : a \text{ is integral over } B\}, \\ \overline{X} &:= \text{Spec } \overline{A}, & Z &:= \text{Spec } A_N. \end{aligned}$$

Thus N/K is the normal closure of L/K .

Proposition 3.8 (Planar finite-cover diagram).

$$Z \xrightarrow[\text{finite}]{\pi} \overline{X} \xrightarrow[\text{finite}]{\bar{\nu}} Y, \quad \nu = \bar{\nu} \circ \pi,$$

and

$$X \xrightarrow[\text{open}]{j} \overline{X}, \quad \bar{\nu} \circ j = f_F.$$

Proof. Since $B \cong \mathbb{C}[u, v]$ is a finitely generated normal $\mathbb{C}$ -algebra, its integral closures in the finite extensions L/K and N/K are finite over B . The inclusion $L \subseteq N$ gives $\overline{A} \subseteq A_N$, so

$$B \subseteq \overline{A} \subseteq A_N.$$

The morphism $\bar{\nu}$ is finite because $\overline{A}$ is a finite B -module. If $a_1, \dots, a_r$ generate A_N over B , then they also generate A_N over $\overline{A}$, since $B \subseteq \overline{A}$. Hence π is finite.

For $a \in \overline{A}$, integrality over $B \subseteq A$ and the normality of $A = \mathbb{C}[x_1, x_2]$ give $a \in A$. Thus

$$B \subseteq \overline{A} \subseteq A,$$

and $\text{Frac}(\overline{A}) = L$: clearing denominators in an algebraic equation over $K = \text{Frac}(B)$ shows that for every $\ell \in L$, some nonzero $b \in B$ makes $b\ell$ integral over B . Thus $\ell = (b\ell)/b \in \text{Frac}(\overline{A})$. The inclusion $\overline{A} \subseteq A$ therefore induces a dominant birational morphism $j: X \rightarrow \overline{X}$.

The morphism j is quasi-finite: because $\bar{\nu} \circ j = f_F$, each fiber of j is contained in a fiber of the quasi-finite morphism f_F . It is separated because it is a morphism between affine schemes. Since $\overline{X}$ is normal, Zariski’s Main Theorem makes j an open immersion. $\square$

3.4 The two planar finiteness targets

Let $\mathfrak{b} \subseteq \overline{A}$ be the radical ideal defining $\partial X = \overline{X} \setminus j(X)$. Because $\overline{X}$ is affine, local cohomology with support in the boundary is

$$H_{\partial X}^1(\overline{X}, \mathcal{O}_{\overline{X}}) = H_{\mathfrak{b}}^1(\overline{A}).$$

Definition 3.9 (The two planar finiteness targets). For a planar Keller map F , define:

1. *PlanarBoundaryCoherence*, denoted $\text{PBC}(F)$, is the assertion that

$$H_{\mathfrak{b}}^1(\overline{A}) \text{ is a finite (equivalently, finitely generated) } \overline{A}\text{-module.}$$

2. *PlanarRamificationRigidity*, denoted $\text{PRR}(F)$, is the assertion that

$$\Omega_{A_N/B} \text{ has finite length as an } A_N\text{-module.}$$

Here finiteness does not repeat the finiteness already proved in Proposition 3.8. The rings $\overline{A}$ and A_N are finite B -modules because they are integral closures in finite field extensions. Consequently $\overline{X}$ and Z are reduced, irreducible affine schemes of finite type over the algebraically closed field $\mathbb{C}$, hence affine varieties in the usual sense. Ordinary higher cohomology of quasi-coherent sheaves on these affine schemes vanishes (the affine-cohomology theorem, or the vanishing half of Serre's affineness criterion) [The26e]. Neither fact implies that the boundary local cohomology $H_{\mathfrak{b}}^1(\overline{A})$ is finite: local cohomology measures sections with poles along the omitted closed set. Similarly, $\Omega_{A_N/B}$ is automatically a finite A_N -module, because A_N is a finite-type B -algebra, but requiring it to have finite *length* is the stronger assertion that its support has dimension zero.

Proposition 3.10 (The boundary-coherence route). *Planar boundary coherence gives the implication chain*

$$\text{PBC}(F) \implies \partial X = \emptyset \implies f_F \text{ is finite étale} \implies F \text{ is a polynomial automorphism.}$$

Proof. Because $\overline{A}$ is a domain and $j(X)$ contains its generic point, $H_{\mathfrak{b}}^0(\overline{A}) = 0$. The local-cohomology sequence for $j(X) \subseteq \overline{X}$, together with affineness of $\overline{X}$, therefore gives

$$0 \longrightarrow \overline{A} \longrightarrow A \longrightarrow H_{\mathfrak{b}}^1(\overline{A}) \longrightarrow 0.$$

Here $\Gamma(X, \mathcal{O}_X) = A$; this is the standard affine open-complement local-cohomology sequence [The26h]. If the last term is a finite $\overline{A}$ -module, then so is A . Thus the open immersion j is finite. Its image is both open and closed; since $\overline{X}$ is integral and $X \neq \emptyset$, it is all of $\overline{X}$. Hence $\partial X = \emptyset$, so $f_F = \bar{\nu}$ is finite. It is already étale by Proposition 3.1, and Proposition 3.2 finishes the proof. $\square$

In particular, $\text{PBC}(F)$ implies $\text{PRR}(F)$: after F is an automorphism one has $A_N = B$, so $\Omega_{A_N/B} = 0$. The converse is not asserted; ramification rigidity eliminates divisorial ramification without first eliminating the entire deleted boundary.

Let $S_F \subseteq Y$ denote the nonproperness set: the complement of the largest open subset $U \subseteq Y$ over which $f_F^{-1}(U) \rightarrow U$ is proper.

Proposition 3.11 (The deleted boundary and properness). *Set-theoretically,*

$$S_F = \bar{\nu}(\partial X).$$

Consequently, the following are equivalent:

$$f_F \text{ is proper,} \quad S_F = \emptyset, \quad \partial X = \emptyset.$$

Under any of these conditions, F is a polynomial automorphism.

Proof. The set $\bar{\nu}(\partial X)$ is closed because $\bar{\nu}$ is finite. Over

$$U = Y \setminus \bar{\nu}(\partial X),$$

the open immersion identifies $f_F^{-1}(U)$ with $\bar{\nu}^{-1}(U)$; hence the restriction of f_F is finite and therefore proper. Thus $S_F \subseteq \bar{\nu}(\partial X)$.

Conversely, suppose that $y \in \bar{\nu}(\partial X)$ and that f_F were proper over a neighborhood of y . Since f_F is quasi-finite, it would be finite after shrinking to an affine neighborhood V of y . The normal finite V -scheme $f_F^{-1}(V)$, with function field L , would then be the normalization of V in L . By uniqueness of normalization it would equal $\bar{\nu}^{-1}(V)$, contrary to the existence of a boundary point above y . Therefore $\bar{\nu}(\partial X) \subseteq S_F$, proving the equality and the three equivalences.

If they hold, $f_F = \bar{\nu}$ is finite and étale. Since $Y \cong \mathbb{A}_{\mathbb{C}}^2$, Theorem 3.30 makes this connected cover trivial, so F is a polynomial automorphism. $\square$

Thus the deleted normalization boundary is not auxiliary: it is precisely the geometric source of nonproperness. For a nonsingular polynomial map of the complex plane, a nonempty S_F is known to be a curve with one point at infinity [Cha04]. The inertia calculations below identify which divisorial part of this nonproperness can support the normal closure.

Here dimension two materially sharpens the normalization problem. Both $\bar{X}$ and Z are normal Noetherian surfaces finite over the regular surface Y . A normal Noetherian scheme of dimension at most two is Cohen–Macaulay [The26a]; since the finite maps have zero-dimensional fibers, miracle flatness makes them finite flat [The26i]. Their coordinate rings are therefore finite locally free B -modules, so the trace, different, and ramification divisor belong to the finite-flat surface setting.

Moreover, $j(X)$ is a dense affine open in the separated Noetherian surface $\bar{X}$. Every irreducible component of $\partial X = \bar{X} \setminus j(X)$ consequently has codimension one [The26b]: a nonempty deleted boundary is a union of curves. Purity of the branch locus rules out ramification supported only in codimension two. Thus the remaining planar question is the following concrete one: can a nonzero ramification divisor on Z have every inertia-moving conjugate center on the deleted boundary curves? The coordinate criterion below gives an exact valuation formulation of this question.

Characteristic zero also makes the ramification divisor of the intermediate normalization explicit. Write

$$R_{\bar{\nu}} = \sum_{C \in \bar{X}^{(1)}} (e(C/\bar{\nu}(C)) - 1)C.$$

The different exponent is $e - 1$ at each divisorial valuation, and the étaleness of f_F gives

$$\text{Supp}(R_{\bar{\nu}}) \subseteq \partial X.$$

The ramification formula moreover identifies $K_{\bar{X}} \sim \bar{\nu}^* K_Y + R_{\bar{\nu}} \sim R_{\bar{\nu}}$.

Proposition 3.12 (Boundary divisor-class rigidity). *Let D be an effective Weil divisor on $\overline{X}$ supported on ∂X . If $[D]$ is torsion in $\text{Cl}(\overline{X})$, then $D = 0$.*

Proof. Choose $m > 0$ and $h \in L^\times$ with

$$\text{div}_{\overline{X}}(h) = mD.$$

The divisor on the right is effective. Since $\overline{X}$ is normal, the height-one valuation criterion for regularity gives $h \in \Gamma(\overline{X}, \mathcal{O}_{\overline{X}}) = \overline{A}$. Its restriction to X has neither zeros nor poles, because D is supported on the deleted boundary. Hence

$$h|_X \in A^\times = \mathbb{C}[x_1, x_2]^\times = \mathbb{C}^\times.$$

The inclusion $\overline{A} \subseteq A$ is injective, so h itself is constant. Therefore $mD = 0$, and hence $D = 0$. $\square$

Corollary 3.13 (Different-class criterion for planar rigidity). *If $[R_{\bar{\nu}}]$ is torsion in $\text{Cl}(\overline{X})$, then F is a polynomial automorphism. In particular, this holds if the different is principal or if $\text{Cl}(\overline{X})$ is torsion.*

Proof. Proposition 3.12 gives $R_{\bar{\nu}} = 0$. Purity of the branch locus then makes the finite map $\bar{\nu}$ étale, and finite-étale rigidity of $\mathbb{A}_{\mathbb{C}}^2$ makes it an isomorphism. Thus $L = K$. Proposition 3.1 gives that A is flat over B , so Corollary 2.17 gives $\text{Obs}(F) = 0$. Theorem 2.15 finishes the proof. $\square$

This isolates a divisor-theoretic rigidity problem specific to the polynomial plane: determine whether the finite-flat trace and different, together with $\overline{A} \subseteq \mathbb{C}[x_1, x_2]$, force $[R_{\bar{\nu}}]$ to be torsion. The proposition proves the consequence; it does not assert this additional class-group statement.

Let

$$\begin{aligned} \mathcal{D}_F &= \left(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, j: X \hookrightarrow \overline{X} \right), & \partial X &= \overline{X} \setminus j(X), \\ G &= \text{Gal}(N/K), & H &= \text{Gal}(N/L). \end{aligned}$$

The abstract conjugate centers now admit a direct two-coordinate test. For a representative $g \in G$ of $\omega = DEgH$, the conjugate affine sheet is generated inside N by

$$A_g = \mathbb{C}[g(x_1), g(x_2)].$$

Proposition 3.14 (Planar coordinate test for a conjugate center). *For $E \in Z^{(1)}$ and $\omega = DEgH \in \mathcal{Q}_E$,*

$$\text{cent}(E, \omega) \in j(X) \iff w_E(g(x_1)) \geq 0 \text{ and } w_E(g(x_2)) \geq 0.$$

Consequently,

$$\text{cent}(E, \omega) \in \partial X \iff \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

Proof. By definition, $u_\omega = (g^{-1}w_E)|_L$, so

$$u_\omega(x_i) = w_E(g(x_i)) \quad (i = 1, 2).$$

The center of u_ω lies on the affine open $X = \text{Spec } \mathbb{C}[x_1, x_2]$ exactly when its valuation ring contains A . Since A is generated by x_1, x_2 , this is equivalent to the two displayed nonnegativity conditions. Negating them gives the boundary criterion. $\square$

The early secant estimate has a precise interpretation on these conjugate generic sheets. For $g \in G$, let

$$\delta_{F,g} := \delta_F(x_1, x_2, g(x_1), g(x_2)) \in N, \quad q_{F,g} := 1 - c^{-1} \delta_{F,g}.$$

Proposition 3.15 (Secant evaluation on conjugate sheets). *For every $g \in G$,*

$$\delta_{F,g}(x_i - g(x_i)) = 0 \quad (i = 1, 2).$$

Consequently,

$$\delta_{F,g} = \begin{cases} c, & g \in H, \\ 0, & g \notin H, \end{cases} \quad q_{F,g} = \begin{cases} 0, & g \in H, \\ 1, & g \notin H. \end{cases}$$

Thus the planar secant projector is exactly the characteristic function of the nonmarked generic conjugate sheets.

Proof. The marked embedding $L \hookrightarrow N$ and its g -conjugate agree on K , so together they define an evaluation map from the generic collision algebra $L \otimes_K L$ to N . Evaluating Proposition 3.4 under this map gives the first display. If $g \in H = \text{Gal}(N/L)$, then $g(x_i) = x_i$ for both coordinates, and diagonal evaluation gives $\delta_{F,g} = c$. If $g \notin H$, then some x_i is moved, since $L = \mathbb{C}(x_1, x_2)$. The corresponding factor $x_i - g(x_i)$ is nonzero in the field N , so the first display forces $\delta_{F,g} = 0$. The formulas for $q_{F,g}$ follow. $\square$

This follows the exponent-one secant bound completely into the normal closure, but also locates its limit. On a nonmarked sheet $\delta_{F,g}$ is the zero function; it is therefore a generic-sheet projector, not a nonzero boundary parameter whose fixed power bounds pole order in $\mathcal{P}_I^{\text{ram}}$. A secant proof of uniform annihilation consequently needs one further transport statement, using more than the determinant alone—for example the adjugate coefficients together with a trace, codifferent, or residue comparison. This is precisely the finite-control step isolated below.

For $E \in \text{Ram}^{(1)}(Z/Y)$ and $\omega \in \mathcal{Q}_E$, Proposition 2.11 gives

$$e(\text{cent}(E, \omega)/\nu(E)) = i_E(\omega).$$

Since $X \rightarrow Y$ is étale by Proposition 3.1, Proposition 2.12 gives

$$i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X.$$

Thus Definition 2.13 specializes to

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

Proposition 3.14 makes this locus fully explicit:

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \forall g \in G, \\ [I_E : I_E \cap gHg^{-1}] > 1 \implies \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0 \end{array} \right\}.$$

The existence of at least one g with nontrivial relative index is automatic from $I_E \neq 1$ and $\text{core}_G(H) = 1$. Thus hidden inertia is not an additional abstract object in the planar proof: it is precisely the possibility that every inertia-moving conjugate coordinate pair acquires a pole at the deleted boundary.

Definition 3.16 (Planar conjugate coverage). Write $\text{Cov}_2(F)$ for the following condition: every $E \in \text{Ram}^{(1)}(Z/Y)$ admits a $g \in G$ such that

$$[I_E : I_E \cap gHg^{-1}] > 1, \quad w_E(g(x_1)) \geq 0, \quad w_E(g(x_2)) \geq 0.$$

Thus some conjugate sheet moved by inertia still has an affine center.

We now apply the boundary principal-parts construction of Section 3.2 to the Galois normalization from Section 2. For $g \in G$, set

$$\overline{A}_g = g(\overline{A}), \quad \overline{X}_g = \text{Spec } \overline{A}_g, \quad X_g = \text{Spec } A_g.$$

The conjugate of j is an open immersion $X_g \hookrightarrow \overline{X}_g$, and the inclusion $\overline{A}_g \subseteq A_N$ gives a finite map $\pi_g : Z \rightarrow \overline{X}_g$. Let

$$U_g := \pi_g^{-1}(X_g) \subseteq Z, \quad D_g := Z \setminus U_g, \quad \mathfrak{j}_g := I_{A_N}(D_g) = \bigcap_{\mathfrak{p} \in D_g} \mathfrak{p}.$$

Thus $\mathfrak{j}_g$ is canonically the radical vanishing ideal of the pulled-back deleted boundary and $D_g = V(\mathfrak{j}_g)$. The construction depends only on gH , since right multiplication by H does not change the conjugate embedding of L .

Proposition 3.17 (Pulled-back boundary dictionary). *For every height-one point E of Z and every $g \in G$,*

$$\eta_E \in U_g \iff \mathfrak{j}_g \not\subseteq \mathfrak{p}_E \iff w_E(g(x_1)) \geq 0 \text{ and } w_E(g(x_2)) \geq 0.$$

Equivalently,

$$\mathfrak{j}_g \subseteq \mathfrak{p}_E \iff \eta_E \in D_g \iff \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

Proof. Because $D_g = V(\mathfrak{j}_g)$, membership $\eta_E \in D_g$ is exactly the containment $\mathfrak{j}_g \subseteq \mathfrak{p}_E$. The point η_E belongs to U_g precisely when its image under π_g is centered on $X_g = \text{Spec } \mathbb{C}[g(x_1), g(x_2)]$. The valuation criterion for a center on an affine model says exactly that both conjugate coordinates have nonnegative valuation. Taking complements gives the second display. $\square$

Proposition 3.18 (Moving-sheet coverage dictionary). *For $E \in \text{Ram}^{(1)}(Z/Y)$, put*

$$\mathfrak{j}_E^{\text{mov}} := \sum_{\substack{g \in G \\ I_E \not\subseteq gHg^{-1}}} \mathfrak{j}_g.$$

Then

$$\exists g \in G, \quad I_E \not\subseteq gHg^{-1}, \quad \eta_E \in U_g \iff \mathfrak{j}_E^{\text{mov}} \not\subseteq \mathfrak{p}_E.$$

Thus eliminating E is exactly the failure of its moving-boundary ideal to be contained in its height-one prime.

Proof. A prime contains a sum of ideals exactly when it contains every summand. Apply Proposition 3.17 to each inertia-moving g . $\square$

For a nontrivial subgroup $I \leq G$, define the moving-boundary ideal and its principal-parts module by

$$\mathfrak{j}_I^{\text{mov}} := \sum_{\substack{g \in G \\ I \not\subseteq gHg^{-1}}} \mathfrak{j}_g, \quad \mathcal{P}_I^{\text{mov}} := H_{\mathfrak{j}_I^{\text{mov}}}^1(A_N).$$

The sum is taken in the common Galois-normalization ring A_N ; this is why the Galois closure enters only at this final specialization.

The same common ring gives a finite-group formulation which does not first choose a ramified divisor. For a subgroup $C \leq G$, let

$$\mathfrak{a}_C := (\sigma(t) - t : \sigma \in C, t \in A_N) \subseteq A_N.$$

This is the scheme-theoretic fixed-locus ideal for the action of C on Z . If $\mathfrak{p}_E \in \text{Spec } A_N$, then

$$\mathfrak{a}_C \subseteq \mathfrak{p}_E \iff C \leq I_E.$$

Indeed, the containment says that every element of C stabilizes $\mathfrak{p}_E$ and acts trivially on its residue field, which is the definition of inertia at E .

Definition 3.19 (Planar boundary separation). Write $\text{Sep}_2(F)$ for the following condition: for every prime-order subgroup $C \leq G$,

$$\text{Spec}^{(1)}(A_N) \cap V(\mathfrak{a}_C + \mathfrak{j}_C^{\text{mov}}) = \emptyset.$$

Equivalently,

$$\forall \mathfrak{p} \in \text{Spec}^{(1)}(A_N), \quad \mathfrak{a}_C \subseteq \mathfrak{p} \implies \mathfrak{j}_C^{\text{mov}} \not\subseteq \mathfrak{p}.$$

In codimension language, every fixed–moving boundary locus $V(\mathfrak{a}_C + \mathfrak{j}_C^{\text{mov}})$ has codimension at least two in the normal surface Z . No assertion is made about isolated fixed boundary points.

Proposition 3.20 (Boundary separation is the divisorial criterion). *For a planar Keller map,*

$$\text{Sep}_2(F) \iff \text{Ram}^{(1)}(Z/Y) = \emptyset.$$

Proof. Assume boundary separation and suppose that $E \in \text{Ram}^{(1)}(Z/Y)$. Choose a subgroup $C \leq I_E$ of prime order; such a subgroup exists by Cauchy’s theorem for the nontrivial finite group I_E . Then $\mathfrak{a}_C \subseteq \mathfrak{p}_E$. If $C \not\subseteq gHg^{-1}$, then $I_E \not\subseteq gHg^{-1}$, so the relative inertia index on the sheet represented by gH is nontrivial. Proposition 2.12 places its center in the deleted boundary, and hence $\mathfrak{j}_g \subseteq \mathfrak{p}_E$. This holds for every sheet moved by C , so

$$\mathfrak{a}_C + \mathfrak{j}_C^{\text{mov}} \subseteq \mathfrak{p}_E,$$

contrary to boundary separation.

Conversely, if boundary separation fails, there are a prime-order $C \leq G$ and a height-one prime $\mathfrak{p}_E$ containing $\mathfrak{a}_C + \mathfrak{j}_C^{\text{mov}}$. The fixed-locus containment gives $C \leq I_E$. Since $C \neq 1$, the point E is ramified, contradicting $\text{Ram}^{(1)}(Z/Y) = \emptyset$. $\square$

Proposition 3.21 (Coverage eliminates a ramified divisor). *Let $E \in \text{Ram}^{(1)}(Z/Y)$. If there is a $g \in G$ such that*

$$I_E \not\subseteq gHg^{-1} \quad \text{and} \quad \eta_E \in U_g,$$

then a contradiction follows. Consequently,

$$\text{Cov}_2(F) \implies \text{Ram}^{(1)}(Z/Y) = \emptyset.$$

Proof. The subgroup condition gives

$$i_E(gH) = [I_E : I_E \cap gHg^{-1}] > 1.$$

On the other hand, $\eta_E \in U_g$ places the conjugate center on the affine sheet X_g . The map $X_g \rightarrow Y$ is the Galois conjugate of the planar Keller map and is therefore étale. Its local ramification index at that center is one. Hence $1 < i_E(gH) = 1$, a contradiction. $\square$

This is the point at which a height-one divisor is eliminated. Purity is not used in Proposition 3.21; it enters only after the proposition has been applied to every ramified height-one point.

Proposition 3.22 (Local-cohomological hidden-inertia detector). *Let $E \in \text{Ram}^{(1)}(Z/Y)$ and put $I = I_E$. Then*

$$E \in \mathcal{H}(\mathcal{D}_F) \iff \mathfrak{j}_I^{\text{mov}} \subseteq \mathfrak{p}_E \iff (\mathcal{P}_I^{\text{mov}})_{\mathfrak{p}_E} \neq 0.$$

Equivalently, conjugate coverage at E is the vanishing of this localization.

Proof. Core-freeness guarantees that the set of inertia-moving sheets is nonempty. The first equivalence follows because a prime contains their sum exactly when it contains every $\mathfrak{j}_g$, which says that every moving center lies in the corresponding deleted boundary. Proposition 3.6 gives the second equivalence. Its nonzero localization is explicitly

$$\text{Frac}((A_N)_{\mathfrak{p}_E}) / (A_N)_{\mathfrak{p}_E}.$$

$\square$

Let $\mathfrak{d}_N = \mathfrak{D}_{A_N/B}$. The ramification-supported part of the preceding module is

$$\mathcal{P}_I^{\text{ram}} := H_{\mathfrak{d}_N}^0(\mathcal{P}_I^{\text{mov}}) \cong H_{\mathfrak{d}_N + \mathfrak{j}_I^{\text{mov}}}^1(A_N).$$

At a ramified E with $I_E = I$, the DVR different is a positive power of the maximal ideal. Hence every principal part is $\mathfrak{d}_N$ -power torsion, and Proposition 3.22 may equivalently use $(\mathcal{P}_I^{\text{ram}})_{\mathfrak{p}_E}$.

Proposition 3.23 (Hidden inertia forces nonfiniteness). *Let $E \in \mathcal{H}(\mathcal{D}_F)$ and put $I = I_E$. Then $\mathcal{P}_I^{\text{ram}}$ is not a finite A_N -module. More precisely,*

$$(\mathcal{P}_I^{\text{ram}})_{\mathfrak{p}_E} \cong \text{Frac}((A_N)_{\mathfrak{p}_E}) / (A_N)_{\mathfrak{p}_E}$$

is not finite over the discrete valuation ring $(A_N)_{\mathfrak{p}_E}$.

Proof. Write $R_E = (A_N)_{\mathfrak{p}_E}$, let $K_E = \text{Frac}(R_E)$, and choose a uniformizer t . Every element of K_E/R_E is killed by some power of t , but there is no uniform exponent: t^{-m-1} is not killed by t^m . If K_E/R_E had finitely many generators, their pole orders would have a common bound, giving such a uniform exponent. Thus the localization is not finite. Since localization preserves finite generation, neither is $\mathcal{P}_I^{\text{ram}}$. $\square$

Proposition 3.24 (Planar finiteness reduction). *Suppose that $\mathcal{P}_I^{\text{ram}}$ is a finite A_N -module for every nontrivial inertia subgroup $I \leq G$ arising at a ramified height-one point. Then $\text{Cov}_2(F)$ holds.*

Proof. If coverage failed, Proposition 3.22 would supply a hidden ramified point E . For $I = I_E$, the finiteness hypothesis would then contradict Proposition 3.23. $\square$

Proposition 3.25 (Uniform annihilation implies planar coverage). *Suppose that for every nontrivial subgroup $I \leq G$ there is an $m_I > 0$ such that*

$$(\mathfrak{d}_N + \mathfrak{j}_I^{\text{mov}})^{m_I} H_{\mathfrak{d}_N + \mathfrak{j}_I^{\text{mov}}}^1(A_N) = 0.$$

Then $\text{Cov}_2(F)$ holds.

Proof. Proposition 3.7 makes each displayed local-cohomology module vanish. If a ramified E existed, take $I = I_E$. Since $\mathfrak{d}_N \subseteq \mathfrak{p}_E$, the vanishing forces $\mathfrak{j}_I^{\text{mov}} \not\subseteq \mathfrak{p}_E$. Hence some inertia-moving U_g contains η_E , which is exactly the required coverage witness. $\square$

The contradiction is therefore finite versus nonfinite. Hidden inertia forces an unbounded tower of local pole classes and hence a nonfinite module; the remaining planar rigidity theorem must force the same module to be finite. There are two natural routes to the latter statement. Since A_N is finite flat over B , its trace dual

$$A_N^\vee = \text{Hom}_B(A_N, B) \cong \mathfrak{D}_{A_N/B}^{-1}$$

is finite; an injective comparison of $\mathcal{P}_I^{\text{ram}}$ with a finite trace-dual or inverse-different quotient would give the required finiteness. Alternatively, the secant annihilator may give a uniform bound on the pole orders directly, after which Proposition 3.7 applies. The present argument establishes neither comparison. Producing one of them from the specific polynomial-plane structure $A = \mathbb{C}[x_1, x_2]$ is the remaining dimension-two task.

Proposition 3.26 (Coverage excludes hidden inertia). *Planar conjugate coverage implies*

$$\mathcal{H}(\mathcal{D}_F) = \emptyset.$$

Proof. Suppose instead that $E \in \mathcal{H}(\mathcal{D}_F)$. Conjugate coverage gives a $g \in G$ with nontrivial relative index and both conjugate coordinates integral. Proposition 3.14 places its center in $j(X)$, whereas Proposition 2.12 places the same center in ∂X . This is impossible. $\square$

3.5 Moving-sheet criteria for planar ramification rigidity

Corollary 3.27 (Planar hidden-inertia identification).

$$\mathcal{H}(\mathcal{D}_F) = \text{Ram}^{(1)}(Z/Y).$$

In particular,

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \iff \text{Ram}^{(1)}(Z/Y) = \emptyset.$$

These conditions are also equivalent to planar conjugate coverage.

Proof. The equality is Proposition 2.14, applied to the planar finite-cover diagram. Conjugate coverage implies empty hidden-inertia locus by Proposition 3.26; conversely, if the ramification locus is empty, conjugate coverage holds vacuously. $\square$

Proposition 3.28 (The ramification-rigidity dictionary). *For a planar Keller map,*

$$\text{PRR}(F) \iff \text{Ram}^{(1)}(Z/Y) = \emptyset \iff \text{Cov}_2(F) \iff \text{Sep}_2(F).$$

Thus moving-sheet coverage and boundary separation are two exact criteria for PlanarRamificationRigidity, not additional global hypotheses.

Proof. The A_N -module $\Omega_{A_N/B}$ is finite, and its generic localization vanishes because N/K is separable. Since A_N is a Noetherian normal domain of dimension two, this module has finite length exactly when it has no height-one point in its support. At a height-one point E , localizing B first at the image of E and then at $\mathfrak{p}_E$ gives the relevant finite local extension; the source local ring is a discrete valuation ring. In characteristic zero its residue extension is separable, so

$$\Omega_{(A_N)_{\mathfrak{p}_E}/B_{\mathfrak{p}_E \cap B}} = 0 \iff I_E = 1.$$

Indeed, vanishing of the localized differential module is the finite-type unramifiedness criterion, and for a finite extension of discrete valuation rings in characteristic zero unramifiedness is equivalent to trivial inertia [The26f, The26c, The26k]. Consequently finite length is equivalent to the absence of divisorial ramification. Proposition 3.21 and vacuity in the unramified case give the equivalence with coverage; the equivalence with separation is Proposition 3.20. $\square$

Equivalently, the premise $\mathcal{H}(\mathcal{D}_F) = \emptyset$ says that if a ramified divisor existed, some inertia-moving conjugate pair $g(x_1), g(x_2)$ would have to remain integral and hence centered on an affine sheet. Planar étaleness says that the same pair must lie at the boundary. This is the boundary contradiction used below; the premise supplies its exclusion, while the preceding results identify exactly what is being excluded.

3.6 The ramification-rigidity endgame

Theorem 3.29 (Purity of the branch locus, specialized). *Let $Z \rightarrow \mathbb{A}_{\mathbb{C}}^2$ be a finite dominant generically separable morphism with Z integral and normal. If $\text{Ram}^{(1)}(Z/\mathbb{A}_{\mathbb{C}}^2) = \emptyset$, then the morphism is étale.*

Proof. At every height-one point of Z , the absence of divisorial ramification gives ramification index one. The corresponding residue field extension is finite and, in characteristic zero, separable. Hence the morphism is unramified at every height-one point. Purity of the branch locus now implies that it is étale everywhere [The26j]. $\square$

Theorem 3.30 (Finite-étale rigidity of the affine plane). *Every connected finite étale cover of $\mathbb{A}_{\mathbb{C}}^2$ is trivial [GR71, Exposé XII, Théorème 5.1].*

Corollary 3.31 (Triviality of the planar normal closure). *If*

$$\text{Ram}^{(1)}(Z/Y) = \emptyset,$$

then

$$N = K \quad \text{and hence} \quad L = K.$$

Numerically,

$$[L : K] = [N : L] = [N : K] = 1.$$

Proof. By Proposition 3.1, $Y \cong \mathbb{A}_{\mathbb{C}}^2$. Since N/K is finite Galois, $\nu: Z \rightarrow Y$ is generically separable; hence Theorem 3.29 makes it finite étale. The inclusion

$$A_N \subset N \text{ is a domain,} \quad \text{so} \quad Z = \operatorname{Spec} A_N \text{ is connected.}$$

Theorem 3.30 therefore makes $\nu: Z \rightarrow Y$ an isomorphism. Taking function fields gives $N = K$. Finally, $K \subseteq L \subseteq N$ implies $L = K$. $\square$

Corollary 3.32 (The ramification-rigidity route). *Planar ramification rigidity gives the implication chain*

$$\operatorname{PRR}(F) \implies Z \longrightarrow Y \text{ is finite étale} \implies N = K \implies L = K \implies q_F = 0 \implies I_R = I_{\Delta}.$$

Proof. Proposition 3.28 first gives $\operatorname{Ram}^{(1)}(Z/Y) = \emptyset$. Purity, Theorem 3.29, makes $Z \rightarrow Y$ finite étale, and Corollary 3.31 gives $N = K$ and $L = K$. Thus $d_F = 1$. Flatness of A over B and Corollary 2.17 give $\operatorname{Obs}(F) = 0$. The abstract off-diagonal projector and collision criteria then give $q_F = 0$ and $I_R = I_{\Delta}$. $\square$

3.7 Vanishing of the off-diagonal factor

Theorem 3.33 (Equivalent boundary criteria for planar vanishing). *For every polynomial map $F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \rightarrow \mathbb{A}_{\mathbb{C}}^2$ satisfying $\det JF = c \in \mathbb{C}^{\times}$, the following conditions are equivalent:*

$$\begin{aligned} \operatorname{PRR}(F) &\iff \operatorname{Sep}_2(F) \iff \operatorname{Cov}_2(F) \\ &\iff \mathcal{H}(\mathcal{D}_F) = \emptyset \iff \operatorname{Ram}^{(1)}(Z/Y) = \emptyset, \end{aligned}$$

and

$$\begin{aligned} \operatorname{Ram}^{(1)}(Z/Y) = \emptyset &\iff d_F = 1 \iff \operatorname{Obs}(F) = 0, \\ \operatorname{Obs}(F) = 0 &\iff q_F = 0 \iff R_F^{\circ} = \emptyset \iff I_R = I_{\Delta}. \end{aligned}$$

Proof. Proposition 3.28 identifies $\operatorname{PRR}(F)$ and $\operatorname{Sep}_2(F)$ with the absence of divisorial ramification. Now assume $\operatorname{Cov}_2(F)$ and suppose, by way of contradiction, that $E \in \operatorname{Ram}^{(1)}(Z/Y)$. Coverage supplies a $g \in G$ satisfying

$$[I_E : I_E \cap gHg^{-1}] > 1, \quad w_E(g(x_1)) \geq 0, \quad w_E(g(x_2)) \geq 0.$$

The valuation inequalities and Proposition 3.14 place the corresponding center in the conjugate affine open, whereas the nontrivial relative index and Proposition 2.12 place it in the corresponding conjugate deleted boundary. This is impossible. Hence $\operatorname{Ram}^{(1)}(Z/Y) = \emptyset$. Conversely, if the ramification locus is empty, coverage holds vacuously. Corollary 3.27 identifies these conditions with $\mathcal{H}(\mathcal{D}_F) = \emptyset$.

Under these conditions, Corollary 3.31 gives $N = L = K$, hence $d_F = 1$, while Proposition 3.1 makes A flat over B . Corollary 2.17 therefore gives $\operatorname{Obs}(F) = 0$. Conversely, if $d_F = 1$, then $L = K$; flatness and Corollary 2.17 again give $\operatorname{Obs}(F) = 0$. If $\operatorname{Obs}(F) = 0$, then $I_R = I_{\Delta}$, so Theorem 2.15 makes F a polynomial automorphism. Hence $B = A$, $K = L = N$, and $\operatorname{Ram}^{(1)}(Z/Y) = \emptyset$.

Corollary 2.4 and Theorem 2.5 state

$$\begin{aligned} R_F^{\circ} = \emptyset &\iff C_F^{\circ} = 0 \iff \operatorname{Obs}(F) = 0 \iff I_R = I_{\Delta}, \\ \operatorname{Obs}(F) = 0 &\iff q_F = 0. \end{aligned}$$

This proves the remaining equivalences. $\square$

Corollary 3.34 (Boundary witness for failure of coverage). *The equivalent conditions of Theorem 3.33 fail if and only if there is an $E \in \text{Ram}^{(1)}(Z/Y)$ such that*

$$\forall g \in G, \quad [I_E : I_E \cap gHg^{-1}] > 1 \implies \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

For such an E , at least one inertia-moving g exists. Thus failure of planar collision vanishing is witnessed by a ramified divisor whose inertia-moving conjugate centers all lie on the deleted boundary; equivalently, the generic degree satisfies $d_F \geq 2$.

Proof. If the equivalent conditions fail, then conjugate coverage fails. Negating Definition 3.16 supplies an $E \in \text{Ram}^{(1)}(Z/Y)$ for which every g of nontrivial relative index fails at least one valuation inequality, giving the displayed condition. Conversely, the displayed condition is incompatible with a coverage witness for E , so conjugate coverage fails. Core-freeness guarantees the existence of at least one g of nontrivial relative index. $\square$

Corollary 3.35 (Finite-group boundary form of the planar Jacobian conjecture). *The following are equivalent:*

1. *Every polynomial self-map of $\mathbb{A}_{\mathbb{C}}^2$ with constant nonzero Jacobian determinant is a polynomial automorphism.*
2. *Every planar Keller map F satisfies $\text{Sep}_2(F)$; equivalently, for every prime-order subgroup $C \leq \text{Gal}(N/K)$,*

$$\text{codim}_Z V(\mathfrak{a}_C + \mathfrak{j}_C^{\text{mov}}) \geq 2.$$

Proof. If the first condition holds, then F is an automorphism, so $K = L = N$, the group G is trivial, and boundary separation is vacuous. Conversely, boundary separation and Theorem 3.33 give $I_R = I_{\Delta}$ for every planar Keller map. The automorphism criterion, Theorem 2.15, then makes every such map a polynomial automorphism. $\square$

The planar section therefore gives an exact codimension-one reformulation. In the first case, conjugate coverage excludes complete boundary-supported inertia orbits and the collision relation collapses to the diagonal. In the second, the collision relation does not collapse and Corollary 3.34 produces a divisorial boundary witness. Asserting that the first case always occurs—or, equivalently, that every planar Keller map satisfies $\text{PRR}(F)$ —is the remaining two-dimensional Jacobian-conjecture statement. The module $\mathcal{P}_I^{\text{ram}}$ packages that witness as an unbounded principal-parts class, so uniform annihilation gives a concrete route from the boundary obstruction back to conjugate coverage and hence to PlanarRamificationRigidity. PlanarBoundaryCoherence is the separate, stronger route that removes the entire normalization boundary at once.

Lean correspondence. The conditional planar chain is collected in `CollisionIdeals/Planar.lean`. The development takes the following established results as explicit inputs: planar Ax–Grothendieck, branch purity, and finite-étale rigidity of $\mathbb{A}_{\mathbb{C}}^2$. It also takes a supplied `PlanarKellerCollisionModelF`. The positive `PlanarHiddenInertia` predicate and the implication from `PlanarConjugateCoverage` to `PlanarNoHiddenInertia` are formalized; constructing the model and proving coverage remain separate formalization tasks. The fixed-locus ideal `fixedLocusIdeal`, the moving-boundary sum `movingBoundaryIdeal`, and the height-one condition `BoundaryIdealData.PlanarBoundarySeparation` are formalized in `CollisionIdeals/Planar/BoundarySeparation.lean`. The Lean predicate quantifies over all nontrivial subgroups; the manuscript’s prime-order formulation is its equivalent Cauchy reduction. The theorem `BoundaryIdealData.ramifiedCodimensionOnePoint_isEmpty` connects that condition to the existing no-ramification endgame. The pulled-back open `pulledBackConjugateAffineOpen`, its closed complement

`pulledBackConjugateBoundary`, and its vanishing ideal `pulledBackConjugateBoundaryIdeal` are constructed canonically; `pulledBackConjugateBoundaryIdeal_le_ramifiedPrime` formalizes the valuation-theoretic containment at a ramified divisor. The exact pointwise coverage assertion and the universal claims $\text{PBC}(F)$ or $\text{PRR}(F)$ remain mathematical hypotheses, not conclusions of the formalization. The local-cohomology target is connected to the downstream endgame by the explicit input `BoundaryCoherenceBridge` and theorem `planarVanishing_of_boundaryCoherence`; the bridge itself is not proved. Likewise, `RamificationRigidityBridge` isolates the unformalized finite-length support step in the weak route. The genuine mathlib local-cohomology objects `BoundaryPrincipalParts`, `RamificationSupportedBoundaryPrincipalParts`, and `HasUniformBoundaryAnnihilator` are defined in `CollisionIdeals/BoundaryPrincipalParts.lean`. The abstract nonfiniteness mechanism for unbounded locally power-torsion principal parts is proved in `CollisionIdeals/Planar/UnboundedPrincipalParts.lean`; finite comparison control and the resulting direct contradiction are formalized in `CollisionIdeals/Planar/FinitePrincipalPartsControl.lean`. The evaluation of the planar secant relation on an arbitrary pair of collision sheets, including its vanishing when a coordinate is moved, is formalized in `CollisionIdeals/Planar/ConjugateSecantEvaluation.lean`. The geometric comparison with sections on an open complement, the height-one DVR detector, the boundary-coherence implication, the finite-length differentials criterion, and the construction of finite control from the planar Keller data remain separate formalization tasks. The properness and divisor-class criteria above are likewise not presently part of the Lean development.

4 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 2 to dimension three and isolate the separable nonnormal cubic case.

Let $S_3 = \text{Perm}(\{1, 2, 3\})$ be the symmetric group on three letters.

Proposition 4.1 (Dimension-three counterexample criterion). *For a polynomial map*

$$F: \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3,$$

consider the following statements:

- (1) $\det JF \in \mathbb{C}^\times$;
- (2) F is not a polynomial automorphism;
- (3) $R_F^\circ \neq \emptyset$.

Statements (2) and (3) are equivalent. Consequently, F is a dimension-three Jacobian-conjecture counterexample exactly when (1) and (3) hold.

Proof. By Corollary 2.4 and Theorem 2.15,

$$R_F^\circ = \emptyset \iff I_R = I_\Delta \iff F \in \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

Negating this equivalence proves $(2) \iff (3)$. Combining this with (1) gives the final assertion. $\square$

4.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3): \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \quad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A),$$

and let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a,$$

be the isomorphism of Theorem 2.6. Suppose that L/K is separable and nonnormal, and choose a power basis with generator α and degree three:

$$L = K(\alpha), \quad [L : K] = 3.$$

If $m_\alpha \in K[T]$ is the minimal polynomial of α , let $h_\alpha \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Let N/K be the normal closure of L/K , containing L through the marked inclusion. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Recall from Section 2.5 that

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \longmapsto g|_L, \quad \text{core}_G(H) = 1.$$

The final equality follows because N is the normal closure of L/K ; equivalently, the action of G on the conjugate embeddings of L is faithful.

For $Y = \text{Spec } B$, the finite-normalization construction of Section 2.5 gives

$$Z = \text{Norm}_N(Y) \longrightarrow \overline{X} = \text{Norm}_L(Y) \longrightarrow Y.$$

If $E \in Z^{(1)}$, with decomposition group D_E and inertia group I_E , then its conjugate centers are indexed by $\mathcal{Q}_E = D_E \backslash G/H$. For $\omega = D_E gH$, Proposition 2.11 states

$$e(\text{cent}(E, \omega)/\nu(E)) = i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

The following proposition is the standard residual-factor description of a separable nonnormal cubic extension. We include its short proof to fix the marked identification with N .

Proposition 4.2 (Cubic residual-factor identification). *The polynomial h_α is irreducible over L , and there is an L -algebra isomorphism*

$$\varphi: L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

Proof. Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

If the quadratic h_α were reducible over L , it would have a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root would be

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus m_α would split over L . Since $L = K(\alpha)$, the extension L/K would then be the splitting field of the separable polynomial m_α , hence normal. This contradicts the assumed nonnormality, so h_α is irreducible.

Let β denote the image of T in $L[T]/(h_\alpha)$. This field contains the three roots

$$\alpha, \quad \beta, \quad -a_2 - \alpha - \beta$$

of m_α , and is generated over L by β . It is therefore the splitting field of m_α , which is the normal closure N of L/K . Choose the resulting L -algebra isomorphism

$$\varphi: L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha: L \otimes_K L \xrightarrow{\sim}_L L \times L[T]/(h_\alpha)$$

be the marked-root decomposition of Theorem 2.7. It satisfies

$$\text{pr}_1 \circ \Phi_\alpha = \mu_L.$$

Composing its second factor with φ , let

$$\Psi = (\text{id}_L \times \varphi) \circ \Phi_\alpha.$$

Since $\text{id}_L \times \varphi$ leaves the first factor unchanged, $\text{pr}_1 \circ \Psi = \mu_L$. □

Theorem 4.3 (Galois group of a nonnormal cubic extension).

$$\text{Gal}(N/K) \cong S_3.$$

Proof. Proposition 4.2 identifies N with the degree-two extension $L[T]/(h_\alpha)$ of L . Hence

$$|H| = [N : L] = 2.$$

The normal-closure identities recalled above give

$$\text{core}_G(H) = 1, \quad [G : H] = [L : K] = 3.$$

Hence the action of G on the three cosets of H gives

$$\rho: G \longrightarrow \text{Perm}(G/H) \cong S_3, \quad \ker \rho = \text{core}_G(H) = 1.$$

Core-freeness therefore makes ρ injective, so

$$|G| \leq |S_3| = 6.$$

The index formula gives

$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$

Thus $|G| = 6$, and the injective homomorphism ρ is an isomorphism $G \cong S_3$. □

4.2 The cubic off-diagonal factor

Retain the cubic data of Section 4.1, and suppose in addition that F is Keller. Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \quad K = \text{Frac}(B) \subseteq L = \text{Frac}(A) = K(\alpha) \subseteq N,$$

where L/K is separable, nonnormal, and of degree three, and N/K is its normal closure. Let

$$\theta: K \otimes_B A \xrightarrow{\sim} L, \quad \lambda \otimes a \mapsto \lambda a.$$

Corollary 4.4 (Generic cubic S_3 collision). *There is a K -algebra isomorphism*

$$\Psi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \quad \text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

It satisfies

$$\Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N, \quad \text{Gal}(N/K) \cong S_3.$$

Consequently,

$$\text{Obs}(F) \neq 0, \quad I_R \subsetneq I_\Delta, \quad R_F^\circ \neq \emptyset, \quad F \notin \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

Proof. Theorem 2.6 gives a K -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 4.2 gives

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

After restricting Ψ from L to K , let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \quad \Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N \neq 0.$$

Suppose that the diagonal-evaluation homomorphism $\bar{\mu}_F: C_F \rightarrow A$ were injective. Since K is a localization of B , it is flat over B , so $1 \otimes \bar{\mu}_F$ would also be injective. Theorem 2.6 shows that θ is an isomorphism. Therefore $\theta \circ (1 \otimes \bar{\mu}_F)$ would be injective, contradicting the nonzero kernel $0 \times N$ above. Hence $\ker(\bar{\mu}_F) \neq 0$, and Proposition 2.2 gives $\text{Obs}(F) \neq 0$. Proposition 2.1 shows that $I_R \subseteq I_\Delta$ strict, and Corollary 2.4 gives $R_F^\circ \neq \emptyset$.

Finally, Theorem 4.3 identifies $\text{Gal}(N/K)$ with S_3 . If F were a polynomial automorphism, Theorem 2.15 would give $\text{Obs}(F) = 0$, contrary to $\text{Obs}(F) \neq 0$. Since F is Keller, it is a dimension-three Jacobian-conjecture counterexample. $\square$

For the same cubic map and field tower, let

$$X = \text{Spec } A, \quad Y = \text{Spec } B, \quad G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Suppose there is a diagram over Y

$$\mathcal{D} = \left(Z = \text{Norm}_N(Y) \xrightarrow{\pi} \bar{X} = \text{Norm}_L(Y) \xrightarrow{\bar{\nu}} Y, j: X \hookrightarrow \bar{X} \right)$$

in which π and $\bar{\nu}$ are finite, j is an open immersion, $\bar{\nu} \circ j = f_F$, and $f_F: X \rightarrow Y$ is étale. Let

$$\partial X = \bar{X} \setminus j(X), \quad G \cong S_3$$

by Theorem 4.3.

Corollary 4.5 (Hidden inertia in the S_3 -normalization).

$$\mathcal{H}(\mathcal{D}) = \text{Ram}^{(1)}(Z/Y).$$

Proof. Apply Proposition 2.14 to the displayed normalization diagram. $\square$

Remark (Ordered roots). Let

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N . For distinct i, j, k ,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \quad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the off-diagonal factor, which marks two distinct roots, is already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/CubicGaloisGroup.lean`,
`CollisionIdeals/ComplexThree/S3Collision.lean`, `CollisionIdeals/ComplexThree/OffDiagonal.lean`,
`CollisionIdeals/ComplexThree/CubicS3.lean`, and
`CollisionIdeals/ComplexThree/JacobianConjecture.lean`. The Lean cubic witness takes the residual-field isomorphism and $H \neq 1$ as explicit inputs; Proposition 4.2 derives them here from nonnormality. Hidden inertia is formalized pointwise at a supplied ramified divisor.

5 Conclusion

The obstruction ideal $\text{Obs}(F)$ vanishes exactly when the scheme-theoretic collision relation $X \times_Y X$ equals the diagonal. In dimension two the two named targets keep the logic separated. `PlanarBoundaryCoherence` removes the whole deleted boundary and makes the original Keller map finite étale. The weaker `PlanarRamificationRigidity` removes only the height-one support of $\Omega_{A_N/B}$; moving-sheet coverage and boundary separation are its exact valuation-theoretic and finite-group criteria. Only then do purity and finite-étale rigidity make the normal closure trivial and force $\text{Obs}(F) = 0$. The image of the deleted boundary is exactly the nonproperness set, and the boundary divisor-class argument gives another sufficient criterion. None of these targets is asserted universally. In the cubic dimension-three class, by contrast, the nonzero generic factor is the ordered-root S_3 -normal closure. Both statements arise from the same collision, off-diagonal, and conjugate-sheet constructions; their point is to locate the obstruction, respectively, at the normalization boundary and on the generic off-diagonal sheet.

Under `PlanarBoundaryCoherence` or `PlanarRamificationRigidity` in dimension two, under the conditions of Section 4 in dimension three, and with $\text{Ram}^{(1)}(Z/Y) \neq \emptyset$ in the cubic normalization, the contrast is

dimension two, conditional	dimension three
PBC(F) or PRR(F)	hidden inertia in the S_3 -cover realized
$\text{Obs}(F) = 0$	$\text{Obs}(F) \neq 0$
$I_R = I_\Delta$	$I_R \subsetneq I_\Delta$
F is a polynomial automorphism	F is not a polynomial automorphism.

Motivating example. The explicit three-dimensional polynomial counterexample announced by Levent Alpöge [Alp26] and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic S_3 comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result is the structural implication from the stated cubic and marked-normal-closure conditions.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

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